

# JAMES FOSTER

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## PROFILE

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Senior lecturer (associate professor) in mathematics at the University of Bath with interests in the numerical analysis of stochastic differential equations and the applications of differential equations to machine learning.

## RESEARCH

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- *Strong Stochastic Flow Maps*. Sam McCallum, Zander Blasingame, Timothy Herschell, Niklas Rindtorff, Alexander Tong and **James Foster** (2026). [arxiv.org/abs/2606.01086](https://arxiv.org/abs/2606.01086).
- *ARCANE: Scalable high-degree cubature formulae for simulating SDEs without Monte Carlo error*. Peter Koepernik, Thomas Coxon and **James Foster** (2026). [arxiv.org/abs/2602.17151](https://arxiv.org/abs/2602.17151).
- *Underdamped Langevin MCMC with third order convergence*. Max Scott, Dáire O’Kane, Andraž Jelinčič and **James Foster** (2026). Journal of Machine Learning Research (JMLR), 27(126):1–63.
- *Approximating the signature of Brownian motion for high order SDE simulation*. **James Foster** (2026). Stochastic Analysis and Applications 2025, Springer.
- *Reversible Deep Equilibrium Models*. Sam McCallum, Kamran Arora and **James Foster** (2025). [arxiv.org/abs/2509.12917](https://arxiv.org/abs/2509.12917).
- *Efficient, Accurate and Stable Gradients for Neural ODEs*. Sam McCallum and **James Foster** (2025). [arxiv.org/abs/2410.11648](https://arxiv.org/abs/2410.11648).
- *On the convergence of adaptive approximations for stochastic differential equations*. **James Foster** and Andraž Jelinčič (2025). [arxiv.org/abs/2311.14201](https://arxiv.org/abs/2311.14201).
- *Single-seed generation of Brownian paths and integrals for adaptive and high order SDE solvers*. Andraž Jelinčič, **James Foster** and Patrick Kidger (2025). [arxiv.org/abs/2405.06464](https://arxiv.org/abs/2405.06464).
- *Generative Modelling of Lévy Area for High Order SDE Simulation*. Andraž Jelinčič, Jiajie Tao, Will Turner, Tom Cass, **James Foster** and Hao Ni (2025). SIAM Journal on Mathematics of Data Science, 7(4):1541–1567.
- *High order splitting methods for SDEs satisfying a commutativity condition*. **James Foster**, Gonçalo dos Reis and Calum Strange (2024). SIAM Journal on Numerical Analysis, 62(1):500–532.
- *Brownian bridge expansions for Lévy area approximations and particular values of the Riemann zeta function*. **James Foster** and Karen Habermann (2023). Combinatorics, Probability and Computing, 32(3):370–397.
- *An asymptotic radius of convergence for the Loewner equation and simulation of SLE traces via splitting*. **James Foster**, Terry Lyons and Vlad Margarint (2022). Journal of Statistical Physics, 189(18).
- *The shifted ODE method for underdamped Langevin MCMC*. **James Foster**, Terry Lyons and Harald Oberhauser (2021). [arxiv.org/abs/2101.03446](https://arxiv.org/abs/2101.03446).
- *Efficient and Accurate Gradients for Neural SDEs*. Patrick Kidger, **James Foster**, Xuechen Li and Terry Lyons. Neural Information Processing Systems 2021.
- *The Signature Kernel is the solution of a Goursat PDE*. Cristopher Salvi, Thomas Cass, **James Foster**, Terry Lyons and Weixin Yang (2021). SIAM Journal on Mathematics of Data Science, 3(3):873–899.
- *Neural SDEs as Infinite-Dimensional GANs*. Patrick Kidger, **James Foster**, Xuechen Li, Harald Oberhauser and Terry Lyons. International Conference on Machine Learning 2021.
- *Neural Rough Differential Equations for Long Time Series*. James Morrill, Cristopher Salvi, Patrick Kidger, **James Foster** and Terry Lyons. International Conference on Machine Learning 2021.
- *Neural Controlled Differential Equations for Irregular Time Series*. Patrick Kidger, James Morrill, **James Foster** and Terry Lyons. Neural Information Processing Systems 2020 (Spotlight).
- *An Optimal Polynomial Approximation of Brownian Motion*. **James Foster**, Terry Lyons and Harald Oberhauser (2020). SIAM Journal on Numerical Analysis, 58(3):1393–1421.

## EDUCATION

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### University of Oxford, Worcester College

*DPhil in Mathematics*

October 2016 – October 2021

*(Thesis published online in 2020)*

- DPhil thesis on the topic of [Numerical approximations for stochastic differential equations](#) can be found online.
- Under the supervision of Prof. Terry Lyons and Prof. Harald Oberhauser, discovered a new relationship between Brownian motion and a class of orthogonal polynomials that has applications to numerical methods for SDEs.
- A Matlab demonstration of this result can be found at [chebfun.org/examples/stats/RandomPolynomials.html](https://chebfun.org/examples/stats/RandomPolynomials.html).
- Awarded a full EPSRC studentship which included a maintenance grant and covered college and tuition fees.

### University of Oxford, St John's College

*MMath in Mathematics*

October 2012 – September 2016

*MMath (First Class Hons), BA (First Class Hons)*

## ACADEMIC PRIZES

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- IMA Leslie Fox Prize for Numerical Analysis 2025 (joint first) awarded for research on SDE splitting methods.
- Presented a [poster](#) in the STEM for Britain 2021 competition (finalist in the mathematical sciences category).
- G-Research PhD Prize in Maths and Data Science 2020 awarded for research on numerical methods for SDEs.

## PROFESSIONAL EXPERIENCE

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### University of Bath, Department of Mathematical Sciences

*Senior Lecturer (associate professor)*

June 2026 – Present

*Bath*

### University of Bath, Department of Mathematical Sciences

*Lecturer (assistant professor)*

August 2022 – May 2026

*Bath*

### University of Oxford, Mathematical Institute

*Postdoctoral Research Associate*

May 2020 – July 2022

*Oxford*

### J.P. Morgan Chase & Co, Global Credit Markets

*Quantitative Research Intern (12 weeks)*

June 2016 – September 2016

*London*

### Credit Suisse International, Fixed Income Division

*Quantitative Summer Analyst (10 weeks)*

June 2015 – August 2015

*London*

### Shell U.K. Limited, Development Engineering Department

*Reservoir Engineer Intern (12 weeks)*

June 2014 – September 2014

*Aberdeen*

### Altera Europe Limited, European Technology Centre

*Intern (1 month)*

September 2013

*High Wycombe*

### Roxar Limited, Software Solutions Division

*Summer Intern (9 weeks)*

June 2013 – August 2013

*Oxford*

## RESEARCH TALKS

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- *From  $1/\sqrt{n}$  to  $1/n$ : Accelerating SDE Simulation with Cubature Formulae.* BIRS Workshop on “New Trends in Rough Analysis: Open Challenges and Applications”, 23/06/2026.
- *Efficient, Accurate and Stable Gradients for Neural Differential Equations.* Beijing Institute of Mathematical Sciences and Applications (BIMSA) Computational Math Seminar, 09/10/2025.
- *High order splitting methods for SDEs.* Workshop “Milstein’s method: 50 years on”, University of Nottingham, 30/06/2025.
- *High order splitting methods for SDEs satisfying a commutativity condition,* IMA Leslie Fox Prize Meeting, University of Strathclyde, 23/06/2025.
- *Efficient, Accurate and Stable Gradients for Neural Differential Equations,* Workshop on “Recent Developments in Theoretical Machine Learning”, Imperial College London, 13/01/2025.
- *Splitting methods and generative modelling for high order SDE simulation.* One World Stochastic Numerics and Inverse Problems seminar, 27/11/2024.
- *An improved Runge-Kutta method for SDEs with additive noise,* Meeting on “Directions in Rough Analysis”, Oberwolfach Research Institute for Mathematics, Germany, 07/11/2024.

- *On the convergence of adaptive approximations for SDEs*, Workshop on “New Impacts of Rough Analysis”, University of Warwick, 25/07/2024.
- *Algebraically reversible solvers for neural differential equations*, Conference on “Signatures of paths and images”, Hotel Lysebu, Oslo, 10/06/2024.
- *Using GANs to improve the simulation of stochastic differential equations*, Joint seminar (“Optimisation and Numerical Analysis” & “Data Science and Computational Statistics”), University of Birmingham, 08/02/2024.
- *High order splitting methods for SDEs*, Stochastic Analysis Seminar, Imperial College London, 30/01/2024.
- *On the convergence of adaptive approximations for SDEs*, CUWB Conference, Playa del Carmen, Mexico, 08/01/2024.
- *High order numerical methods for SDEs*, North-East and Midlands Stochastic Analysis (NEMSA) Seminar, Oxford, 26/09/2023.
- *Markov Chain Cubature for Bayesian Inference*, Minisymposium on “Methodological advancement in rough paths and data science”, International Congress on Industrial and Applied Mathematics (ICIAM), 24/08/2023.
- *High order splitting methods for SDEs satisfying a commutativity condition*, Invited session on “Numerical methods for SDEs” at the Conference on Stochastic Processes and Applications, Lisbon, Portugal, 24/07/2023.
- *Markov Chain Cubature for Bayesian Inference*, BIRS Workshop on “New interfaces of Stochastic Analysis and Rough Paths”, 08/09/2022.
- *Applications of high order SDE solvers in machine learning*, Dagstuhl Seminar on “Differential Equations and Continuous-Time Deep Learning”, Schloss Dagstuhl, Wadern, Germany, 16/08/2022.
- *Neural Stochastic Differential Equations for Time Series Modelling*, “ICMS at Oxford” workshop on Advances in  $N$ -body Computations, 11/04/2022.
- *A high order method for underdamped Langevin MCMC*, International Conference on Monte Carlo Methods and Applications, 18/08/2021.
- *Log-signatures and Neural Rough Differential Equations*, ICERM Workshop on Applications of Rough Paths: Computational Signatures and Data Science, 7/03/2021.
- *Understanding randomness with polynomials*, STEM for Britain – Mathematical sciences category, 03/03/2021.

## TEACHING EXPERIENCE (BATH)

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- Colecturer (with Eike Mueller) for a first-year course on “Programming for Mathematics” (2025 – present).
- Lecturer (or unit convenor) for a second-year undergraduate course on “Numerical Analysis” (2023 – present).
- Lecturer (or unit convenor) for a new graduate course on “Applied Machine Learning” (2024-25).
- Lecturer (or unit convenor) for a SAMBa PhD course on “Applied Stochastic Differential Equations” (2023-24).
- Computer lab tutor for first-year undergraduate courses on Python programming (2023 – present).
- Delivered a lecture and coursework on “Brownian motion and Stochastic Differential Equations” as part of an undergraduate course called “Connections”, which aims to introduce students to more advanced topics (2023).
- Supervisor for SAMBa Research Group Project on “Exploring applications of smarter subsampling” (2025-26).
- Supervisor for SAMBa Interdisciplinary Research Project on “Multiscale Flow in Porous Media” (2023-24).
- Supervisor on eight separate reading courses on SDEs and machine learning for SAMBa PhD students (2022-24).
- Contributed to exam and coursework marking for the undergraduate course “Numerical analysis” (2022-26).
- Seminar leader for “Foundations”, an introductory course on university mathematics for students (2022-23).

## TEACHING EXPERIENCE (OXFORD)

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- Contributed to the exam marking for the undergraduate course “Probability, Measure and Martinagles” (2022).
- Tutor for the undergraduate courses: Information Theory (2021, 2022) and Communication Theory (2018).
- Member of selection committee for undergraduate admissions at Worcester College (2017, 2019, 2020).
- Graduate lecturer at Worcester College, tutoring Linear Algebra and Real Analysis (2017 – 2018).
- Teaching assistant for courses on Financial Derivatives (2017) and Communication Theory (2016, 2017).

## PHD STUDENTS

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- Harry Semple (Bath). Primary supervisor with Eike Mueller as second supervisor (2026 – present).
- Timothy Herschell (Bath). Primary supervisor with Tony Shardlow as second supervisor (2025 – present).
- Samuel McCallum (Bath). Primary supervisor with Neill Campbell as second supervisor (2024 – present).
- Maximilian Scott (Bath). Primary supervisor with Chris Budd as second supervisor (2024 – present).
- Andraž Jelinčič (Bath). Primary supervisor with Neill Campbell as second supervisor (2023 – present).
- Peter Crew (Bath). Second supervisor with Avi Mayorcas as the primary supervisor (2025 – present).
- Dáire O’Kane (Bath). Second supervisor with Avi Mayorcas as the primary supervisor and Chris Budd as third supervisor (2023 – present).
- Calum Strange (Edinburgh). Second supervisor with Gonçalo dos Reis as the primary supervisor (2021-2023). The first half of Calum’s thesis is on the topic of [Path-based splitting methods for SDEs](#) and is available online.
- Aaron Fordonnell (Bath, Engineering). Second supervisor with Benjamin Metcalfe as the primary supervisor and Matthew Nunes as third supervisor (2023-2025).

## SUPERVISION OF STUDENT PROJECTS

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- Bradley Earley (Bath, BSc). Summer project on “Second-order exponential integrators for ODEs” (2026).
- Jacob Adriano (Bath, MEng). Summer project on “Cubature vs Monte Carlo for optimal control” (2026).
- Nathan Evans (Bath, BSc). Summer project on the “Structured Noise to Reduce Deep BSDE Training Loss”, jointly funded by the LMS and the University of Bath. The [report](#) and Python code are available [online](#) (2025).
- William Warren (Bath, BSc). Bsc project on “Numerical methods for stochastic differential equations” (2025).
- Dylan Nimmo (Bath, MMath). Masters project on “Deep BSDE method for high-dimensional PDEs” (2024-25).
- William Warren (Bath, BSc). Summer project on “Using recombination for creating cubature formulae” (2024).
- Samuel McCallum (Bath, PhD). Reading course on “Parameter inference for SDEs from time series data” (2024).
- Kamran Arora (Bath, PhD). Reading course on “High order splitting methods for SDEs and SPDEs” (2024).
- Yasir Abdi (Bath, PhD). Reading course on “Interacting particle optimization for sampling problems” (2024).
- Julia Zysko (Bath, MMath). Year-long masters project on “Numerical approximations for SDEs” (2023-24).
- Joshua Abbs (Bath, MSc). Masters project on “Cutting-Edge Audio Deepfake Generation Techniques” (2023).
- Sébastien Vol (Telecom SudParis, MEng). Summer internship on “SDE Cubature for Derivative Pricing” (2023).
- Pablo Arratia Lopez (Bath, PhD). Reading course on “Parameter inference for SDEs from snapshot data” (2023).
- Daniel Burrows (Bath, PhD). Project on “Clustering algorithms for SDE-based particle methods” (2022-23).
- Andraž Jelinčič (Oxford, MMathCompSci). Masters project on “GAN-based Lévy area Simulation” (2022-23).
- Patrick Fahy (Bath, PhD). Reading course on “High order numerical methods for additive noise SDEs” (2022).
- Veronika Chronholm (Bath, PhD). Reading course on “Multilevel Monte Carlo for high order methods” (2022).
- Guannan Chen (Bath, PhD). Reading course on “Neural Differential Equations” with Lisa Kreusser (2022).
- Pablo Arratia Lopez (Bath, PhD). Reading course on “Neural Differential Equations” with Pranav Singh (2022).
- Akira Terada (Edinburgh, MSc). Masters project on “High order Antithetic Multilevel Monte Carlo” (2022).
- Katie Zhang (Oxford, MMath). Summer project on “Cubature for SDEs” and funded by Marshall Wace (2022).
- Matthew Young (UCL, MSc). Summer project on “Cubature for SDEs” as part of UNIQ+ Internship (2022).

## ORGANISATION AND SERVICE

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- Organiser of the Numerical Analysis and Data Science seminar at the University of Bath (2024 – present).
- Lead organiser of an ICMS Strategic Workshop on [New Directions for Stochastic Differential Equations and Machine Learning](#) at the Bayes Centre, University of Edinburgh (3-7 June 2024).
- Member of organising committee for a conference on [Modern Topics in Stochastic Analysis and Applications](#) at Imperial College London (22-26 April 2024). Editor for the subsequent conference proceedings cited below:

*Stochastic Analysis and Applications 2025: In Honour of Terry Lyons*. Editors: Dan Crisan, Ilya Chevyrev, Thomas Cass, **James Foster**, Christian Litterer and Cristopher Salvi. Springer (2026).

- Member of organising committee for a workshop on [Dynamics, Data and Deep learning](#) (25-26 March 2024).
- Internal PhD examiner at the University of Bath for Ivan Cheltsov – who is expected to graduate in 2026/27.
- Reviewer of a proposal for a Leverhulme Trust Research Project.
- Reviewer for the International Centre for Mathematical Sciences (ICMS) “Research in Groups” programme.
- Reviewer for the SIAM Journal on Numerical Analysis, the SIAM Journal on Mathematics of Data Science and the SIAM Journal on Control and Optimization.
- Reviewer for the SIAM Journal on Financial Mathematics and the Risk Journal on Computational Finance.
- Reviewer for the International Conference on Neural Information Processing Systems (NeurIPS), 2021 & 2022. Received a NeurIPS Outstanding Reviewer Award in 2021 (given to the top 8% of reviewers).
- Reviewer for the International Conference on Learning Representations (ICLR), 2022.